

Activity Guide — senior / module-11-multiplication-near-base

IKS Curriculum

Table of Contents

Activity 1 — Method Speed Race

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Duration:** 20 min

Purpose

Reinforce the day-by-day concepts of Multiplication Near a Power of 10 — Nikhilam through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

20 problem cards, stopwatches, two columns on whiteboard

What students do

Split class in two: one team solves with traditional long multiplication, the other with Nikhilam. Race on identical problems. Discuss which problems favour which method.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (10 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.

- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Project Brief

Project Brief — Personal Mental-Math Toolkit

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior

What you will do

Each student compiles a one-page toolkit: when to use $\times 11$, when Nikhilam multiplication, when traditional long-mult. Include 3 worked examples for each method.

Why

This project asks you to apply concepts from this module to your own life or a real situation. It is graded on the rubric in `rubric.md`.

Deliverables

- **One-page project plan** (Day 2 of the module): problem, sources, deliverable, timeline.
- **Daily journal entries** (one per day, starting Day 6): what you did, what worked, what didn't.
- **Final artefact** (a poster / model / digital deliverable).
- **Written reflection** (500 words (Senior)): what you learned, what surprised you, what you're still uncertain about.
- **Presentation** (5-minute share with the class).

Timeline

- **Day 1–2:** scope and plan.
- **Day 3–4:** research.
- **Day 5:** mid-project critique.
- **Day 6–7:** build / make.
- **Day 8:** document.
- **Day 9:** rehearse.
- **Day 10:** share day.

Sources

You must cite at least 3 sources. Use the form: *Author, Title, year*. (See `../sources.md` for examples.)

Safety

- Any project involving plants / food / outdoor work must be supervised.
- **No herbal medicine recommendations.** Modules 10 and 4 give awareness; therapeutic decisions belong to qualified practitioners.

Grading

See `rubric.md`. The project is worth the same weight as the summative quiz.